Understanding Tourist Experience in NYC

Final Project Report for the course

Realtime and Big Data Analytics

by

Shivam Balikondwar New York University

Aygun Najafova New York University

Era Sarda New York University

Debdeep Naha New York University

under the guidance of

Instructor
Prof.Yang Tang

to the

Courant Institute

NEW YORK UNIVERSITY
NEW YORK - 10012, NY

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Abstract

Large-scale urban analytics require the integration and processing of heterogeneous data sources that vary in structure, volume, and update frequency. In this work, we present a scalable data engineering and analytics framework for characterizing New York City neighborhoods at the zip code level using Apache Hive on top of HDFS. Multiple public datasets, including restaurant inspection records, hotel review data, crime statistics, and subway ridership logs, are ingested, stored, and processed using distributed query execution and MapReduce-style aggregations.

We design a unified Hive-based schema that enables efficient joins and neighborhood-level summarization across billions of records. Using HiveQL, we derive aggregated indicators capturing food quality, lodging quality, public safety, transit accessibility, cuisine diversity, and temporal mobility patterns. In addition, ridership-based behavioral metrics, including a Leisure Intensity Score and a Nightlife Activity Index, are computed to distinguish commuter-centric districts from leisure- and nightlife-oriented neighborhoods. Descriptive visual analysis of the aggregated outputs reveals systematic spatial patterns across boroughs and dominant cuisine types. This study demonstrates how Hive and distributed storage can be effectively leveraged to support scalable, multi-dimensional urban neighborhood characterization for tourism and urban analytics applications.

Contents

Background

With the growing tourism industry worldwide, data is accumulating daily through hotel and restaurant reviews, social media updates, local public transportation logs, crime incident reports,and data from aviation and railway industries, etc. Various studies are utilizing big data to analyze tourism sentiments, convenience, and preferences for visiting different destinations. However, many of these studies focus on single-scope datasets, which may miss the nuanced trade-offs that tourists face.

For example, a hotel with excellent reviews might be predominantly visited by groups with more than one person, overlooking safety concerns and the low foot traffic in nearby metro stations. If a solo traveller books the hotel based solely on the favorable reviews, they could end up having less-than-ideal experience due to the limited traffic in nearby metro stations. Similarly, an area with high transportation accessibility and foot traffic may have restaurants with lower ratings on health inspection ratings.

This issue is particularly relevant in large metropolitan cities like New York City, where a wide variety of options and area dynamics come into play. Each of these factors contributes to the overall travel experience and the city's reputation as a global destination.

We developed an integrated analytics pipeline that uses the four main large-scale public datasets: hotel reviews (indicating traveler stay satisfaction and sentiment), NYC Restaurant Inspection Results (assessing food quality and cleanliness), MTA Subway Hourly Ridership (measuring transportation accessibility, hourly foot traffic, which served as a proxy for perceived safety during non-rush hours), and NYPD Arrest Data (providing an indirect proxy for local safety and security).

By analyzing these datasets, we aimed to determine how several factors influence tourist satisfaction and attraction. The findings help us understand the interplay between city dynamics and traveler decision-making. This will support (1) tourists in making informed decisions while visiting NYC, and (2) the development of data-driven urban design strategies to improve NYC as a tourist destination.

About the Dataset

This study draws on four primary datasets that collectively capture public health, mobility, safety, and user-generated perception data within New York City. These datasets were selected to provide complementary perspectives on tourist perceptions and satisfaction, and all are well-suited for large-scale, data-driven analysis due to their richness, temporal coverage, and geographic detail.

The first dataset used in this study is the NYC Restaurant Inspection dataset [1], which is published by the City of New York Department of Health and Mental Hygiene (DOHMH) through the NYC Open Data portal. This dataset contains detailed records of restaurant inspections, including a unique restaurant identifier (CAMIS), establishment information (such as name, address, borough, ZIP code, and cuisine type), inspection outcomes (inspection date, violation codes and descriptions, critical flags, inspection scores, grades, and actions taken), and spatial attributes (latitude, longitude, community board, council district, and census tract). The dataset is approximately 150 MB in size and contains around 291,000 rows. It is publicly available for download and does not require any special approval or restricted access.

The second dataset consists of hotel review data collected from TripAdvisor [5]. This dataset, shared publicly via Kaggle, contains user-generated reviews of hotels, including review text, ratings, and associated metadata such as hotel identifiers, review dates, and reviewer locations. The dataset was originally scraped from the TripAdvisor platform and made available for research purposes. It is substantially larger than the restaurant inspection dataset, with a size of approximately 1 GB and roughly 631,300 review records. Access is provided through Kaggle, and no additional authorization beyond standard platform usage is required.

The third dataset analyzed in this study is the NYC Crime Complaint Data [3], which is provided by the New York City Police Department (NYPD) via the NYC Open Data portal. This dataset includes over 2 million complaint records, amounting to approximately 1.45 GB of data, and contains around 36 features per record. These features include complaint identifiers, dates and times of occurrence, borough and precinct information, offense types and descriptors, law category codes, and geographic coordinates. The dataset offers fine-grained spatial and temporal detail on reported crime incidents across the city and is fully accessible as public open data without the need for special permissions.

The fourth dataset is the MTA Subway Hourly Ridership dataset [2], which is published by the Metropolitan Transportation Authority through the New York State Open Data portal (Data.NY.gov). This dataset provides hourly ridership counts across subway station complexes from 2020 through 2024. It contains 12 primary columns, including station complex identifiers, timestamps, fare payment class, ridership counts, borough information, and geographic coordinates. Due to its high temporal resolution and long time span, the dataset is particularly large, exceeding 1 GB in size and comprising approximately 121 million rows. Like the other city-owned datasets, it is publicly available and can be downloaded without special approval.

For the purposes of this study, all datasets are treated as static snapshots. Each dataset is downloaded once, stored locally on the computing cluster, and analyzed without continuous updates or streaming ingestion. Access to the datasets is achieved through open data portals or third-party platforms, and no special institutional approval is required. Importantly, each dataset consists of individual-level records—such as a single restaurant inspection, crime complaint, hotel review, or hourly ridership measurement—and includes spatial attributes such as latitude and longitude, ZIP code, or borough. These shared geographic characteristics enable cross-dataset alignment and support the spatial analyses central to this work.

Implementation

3.1 High-level Implementation of the System

3.1.1 System Architecture

The implementation follows a layered architecture that separates data ingestion, processing, storage, and presentation concerns (Figure 3.1). This modular design enables independent scaling and maintenance of each component while ensuring data flows efficiently from raw sources to end-user visualizations.

Storage Layer. The foundation of the system utilizes HDFS (Hadoop Distributed File System) for distributed storage of both raw and processed data. Raw data from all sources is initially stored in a `/raw/` directory structure, providing a persistent landing zone for ingested datasets. This distributed storage approach ensures fault tolerance through data replication and enables parallel access for downstream processing operations.

Processing Layer. The core processing pipeline implements a MapReduce paradigm divided into three sequential stages. The first stage performs data profiling and quality checks, systematically examining datasets for completeness, consistency, and structural integrity. This profiling generates metadata about data distributions, missing values, and potential anomalies that inform subsequent processing decisions.

The second stage executes data cleaning and standardization operations. MapReduce jobs apply transformation rules to address quality issues identified during profiling, including missing value handling, format standardization, categorical variable normalization, and logical consistency validation. The cleaned datasets are persisted to HDFS in a `/cleaned/` directory.

The third stage implements data integration, filtering, and aggregation logic. This phase combines information from multiple sources where appropriate, applies domain-specific filtering rules, and performs necessary aggregations to create a unified analytical dataset. The integrated results are stored in the `/integrated/` directory, representing the final processed state ready for querying.

Data Warehousing Layer. Apache Hive provides the data warehousing and query interface layer. External tables are created to reference the integrated data stored in HDFS, allowing users to interact with the distributed datasets using familiar SQL-like syntax. This abstraction layer shields analysts from the underlying complexity of distributed data processing while maintaining the scalability benefits of the Hadoop ecosystem.

Visualization Layer. The presentation tier leverages Tableau to create interactive dashboards that connect directly to the Hive data warehouse. This integration can enable real-time querying of processed data while providing intuitive visual interfaces for exploratory analysis. The visualization layer translates analytical queries into efficient Hive operations, ensuring responsive performance for interactive exploration and insight generation.

This architecture provides a scalable, maintainable pipeline that transforms raw heterogeneous data into actionable insights through systematic processing and accessible visualization interfaces.

Fig. 3.1: System Architecture showing data flow from sources through HDFS storage, MapReduce processing, Apache Hive querying, to visualization tools.

3.1.2 Data Integration Flow

The data integration process employs a MapReduce workflow to consolidate four heterogeneous datasets into a unified analytical framework (Figure 3.2). The pipeline systematically filters, extracts, and aggregates location-based information to generate composite metrics representing different dimensions of urban tourism quality.

Processing Pipeline. The integration workflow executes three sequential MapReduce operations. The first operation filters entries to retain only New York City records, establishing geographic consistency across all input sources. This step standardizes the spatial scope and eliminates records outside the study area.

The second operation extracts analytical attributes from each dataset based on their respective domains. This extraction transforms raw records into structured features suitable for spatial integration, isolating relevant measurements such as ratings, grades, ridership volumes, and incident counts alongside their geographic identifiers.

The third operation performs location-based aggregation using geographic coordinates as the join key. This spatial consolidation creates unified location profiles by combining information from all sources at the neighborhood level, enabling cross-domain analysis within defined geographic boundaries.

Feature Generation. The aggregated data produces three composite indicators. Hotel Accessibility measures combine proximity to public transportation infrastructure, quantifying transit access for hospitality establishments. Dining Quality metrics aggregate restaurant inspection grades within geographic areas to assess food service standards at the neighborhood level. Safety Scores incorporate spatial crime density calculations to generate location-based safety assessments.

Index Synthesis. The final stage combines the three composite features into an Overall Tourist Satisfaction Index through weighted aggregation. This unified metric captures multiple dimensions of destination quality in a single quantitative measure, enabling comparative analysis across neighborhoods and temporal monitoring of tourism infrastructure conditions. The integration demonstrates how distributed processing efficiently combines large-scale datasets to generate derived analytics beyond the scope of traditional single-machine computation.

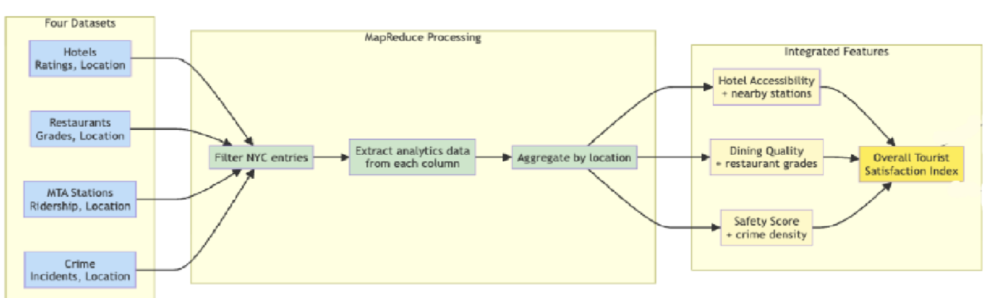

Fig. 3.2: Data Integration Flow showing how MapReduce operations create inte- grated features for analysis.

3.2 Data Cleaning and Pre-processing

Given the heterogeneity, scale, and varied data formats of the datasets used in this study, a comprehensive data cleaning and preprocessing pipeline was essential to ensure consistency, reliability, and analytical validity. The datasets span multiple domains and were sourced from different platforms with distinct schemas and quality challenges. To address these challenges, a unified preprocessing strategy was adopted using distributed data processing techniques on a Hadoop-based infrastructure. This section describes the preprocessing steps applied across all datasets, with dataset-specific adaptations where necessary.

NYPD Arrests Data A.1. The NYPD Arrests dataset, obtained from the NYC Open Data portal, comprises approximately 5.9 million arrest records spanning from 2006 to 2024. Due to its size and structural variability, preprocessing was implemented using a MapReduce pipeline on a Hadoop Dataproc cluster. Initial ingestion involved schema validation and robust CSV parsing to handle escaped characters and quoted fields. Records with malformed structures or incorrect column counts were discarded to ensure schema consistency.

Temporal attributes were standardized into a uniform datetime format, and records with invalid or missing arrest dates were removed. To focus the analysis on contemporary crime patterns, only arrests occurring on or after January 1, 2015 were retained. Categorical variables such as offense type and borough codes were normalized to their semantic representations, while numeric code-based attributes were validated and cast into appropriate data types. Demographic attributes, particularly age group information, were decomposed into minimum and maximum age bounds to enable finer-grained analysis.

Geospatial preprocessing was applied by converting latitude and longitude coordinates into ZIP codes through a point-in-polygon spatial lookup against official New York City boundaries. Once validated ZIP codes were derived, raw geographic coordinates were removed. The final output consisted of a cleaned arrest-level dataset and an aggregated statistics file summarizing crime counts by borough, year, ZIP code, age group, and gender.

NYC Restaurant Inspection Data A.3. Complementing the crime dataset, the NYC Restaurant Inspection dataset contains over 290,000 inspection records with attributes describing inspection outcomes, violations, and restaurant metadata. Preprocessing focused on improving data completeness and temporal relevance. Records missing critical fields such as inspection scores, ZIP codes, violation codes, or inspection dates were removed, as these attributes are essential for spatial and quality-based analysis. Additionally, inspections conducted prior to January 1, 2015 were excluded to align temporally with other datasets.

Using a MapReduce-based pipeline, records were parsed with a schema-aware CSV parser, and textual fields were standardized through trimming and null normalization. Numerical inspection scores were validated and cast into integer format. Inspection grades, which were frequently missing, were derived programmatically from inspection scores based on official NYC Department of Health guidelines. To reduce dimensionality, only analytically relevant attributes were retained, while extraneous metadata and raw geographic coordinates were removed. The reducer phase emitted a consolidated, schema-consistent CSV file suitable for downstream querying and visualization.

MTA Subway Hourly Ridership Data A.2. The MTA Subway Hourly Ridership dataset contains approximately 36.7 million records from 2020 to 2024 and exceeds 6 GB in size. Due to its scale, the raw data was partitioned and ingested into HDFS before preprocessing using Hadoop Streaming with Python-based MapReduce jobs.

Data cleaning addressed formatting inconsistencies in numeric and temporal attributes. Thousands separators were removed from ridership counts, timestamps were normalized to ISO-8601 format, and derived temporal features such as hour of day, day of week, and year were extracted. Duplicate records were identified using an MD5 hash generated from a composite key consisting of timestamp, station identifier, payment method, and fare class category. Reducers ensured record-level uniqueness by retaining only the first occurrence of each hash.

A subsequent profiling job assessed data completeness and distribution at scale. To mitigate shuffle-phase bottlenecks caused by high-cardinality attributes, an in-memory combiner was introduced, reducing intermediate data volume by over 99%. The final cleaned dataset achieved a completeness rate exceeding 99.99% and was written to HDFS in a standardized CSV format for integration with other datasets.

Hotel Reviews Data A.4. Unlike the tabular datasets, this data was stored as nested JSON objects, with each hotel record containing multiple reviews and associated metadata. Preprocessing therefore focused on structural transformation and normalization. Nested rating attributes, such as cleanliness, service, location, and overall experience, were flattened into individual numerical columns. Address information was decomposed into region, street address, locality, and ZIP code, and a consolidated full-address field was constructed to support spatial analysis.

Temporal attributes posed additional challenges, as stay dates were represented in heterogeneous formats. These were normalized to ISO-8601 format by standardizing the day component, and missing stay dates were imputed using review dates.

The pre-cleaned JSON data was then processed through a MapReduce pipeline, where each review was validated for completeness and relevance. The mapper emitted both a cleaned review-level dataset and a statistics dataset containing numerical ratings for aggregation. To ensure geographic and temporal consistency with othet datasets, only reviews associated with

New York City hotels and stay dates between 2020 and 2024 were retained. Additionally, any records which had missing text review as well as missing ratings for all the attributes: service, cleanliness, overall, rooms, location, value, and sleep quality, were removed. Any personal identification information of the reviewer like name and phone number were also dropped. These transformations resulted in a structured and scalable dataset suitable for sentiment analysis, hotel ranking, and spatiotemporal trend analysis.

Across all datasets, the preprocessing pipeline emphasized schema consistency, temporal alignment, spatial enrichment, and noise reduction. By leveraging distributed processing and dataset-specific validation strategies, the final cleaned datasets provide a robust and interoperable foundation for subsequent analytical tasks.

3.3 Data analysis using Hive

Our analysis is based on spatial attributes, specifically zip code and borough, to integrate information from the heterogeneous datasets. The datasets were joined using HiveQL, with the five-digit NYC zip code serving as the primary join key across all tables.

The selected attributes are intended to provide a high-level, consolidated view of neighborhood characteristics that are relevant to tourists and short-term visitors. At this stage, rather than performing downstream statistical or predictive analysis, the objective was to construct a unified table that enables efficient querying and exploration of key neighborhood indicators at the zip-code level.

Table 3.1 illustrates the schema of the primary integrated table produced through this process. This table functions as a core search and aggregation layer and contains pre-computed summary metrics such as average hotel rating (*avg_hotel_rating*), percentage of restaurants receiving an A health grade (*pct_grade_a*), total number of arrests (*total_arrests*), total subway ridership (*total_ridership*), and cuisine diversity (*cuisine_diversity*). These attributes collectively offer a concise snapshot of neighborhood conditions for each zip code.

At this stage, no inferential or comparative analysis was performed on the integrated table. Instead, it was designed to serve as a reusable, high-level data layer that could support subsequent spatial analyses, exploratory queries, and application-driven use cases that require neighborhood-level summaries.

Behavioral Indicators of Neighborhood Activity. To further characterize neighborhood usage patterns, we derived two behavioral indicators from subway ridership data: a Leisure Intensity Score and a Nightlife Activity Index. The Leisure Intensity Score, defined as the ratio of cumulative weekend ridership to cumulative weekday ridership, was used to distinguish commuter-oriented areas from leisure-focused neighborhoods. Our observations

Column	Description
`zipcode`	5-Digit NYC Zipcode (The unique key).
`borough`	Borough name (Manhattan, Bronx, etc.).
`station_count`	Number of unique subway stations in that zipcode.
`hotel_count`	Number of unique hotels listed.
`avg_hotel_rating`	Average customer rating (1–5 scale).
`restaurant_count`	Number of restaurants inspected.
`pct_grade_a`	Percentage of restaurants with an ‘A’ health grade.
`total_arrests`	Total number of arrests in 2024.
`total_ridership`	Total annual subway entries/exits.
`cuisine_diversity`	Count of unique cuisine types.
`dominant_cuisine`	Most common food type.

Table 3.1: Schema of the primary integrated table created using HiveQL.

indicate that zip codes associated with business districts, such as the Financial District in Lower Manhattan, consistently exhibited lower Leisure Intensity Scores (below 0.20), reflecting strong weekday commuter dominance. In contrast, neighborhoods known for cultural, dining, and recreational activities, uch as Williamsburg and portions of Midtown and Lower Manhattan, demonstrated higher scores (above 0.35), suggesting a greater concentration of weekend and leisure-driven travel.

Complementing this measure, the Nightlife Activity Index was calculated as the proportion of ridership occurring between 10 PM and 4 AM relative to total annual ridership. Elevated Nightlife Index values were observed in areas such as the Meatpacking District and other central Manhattan neighborhoods, indicating sustained late-night activity consistent with entertainment and nightlife offerings. Conversely, predominantly residential areas, particularly in Queens, exhibited substantially lower late-night ridership shares, reflecting quieter nighttime mobility patterns. Together, these findings highlight how temporal ridership patterns can effectively differentiate commuter hubs, leisure-oriented destinations, and nightlife centers, providing additional behavioral context for neighborhood-level characterization.

Observations and Trends

As per our earlier datasets and the combined analysis, we found some interesting aspects to study the trends in NYC.

4.1 Study of data distribution by zipcode

In this study, we executed the aggregate queries on HiveQL to study the distributions of each record in each dataset as per the zipcode.

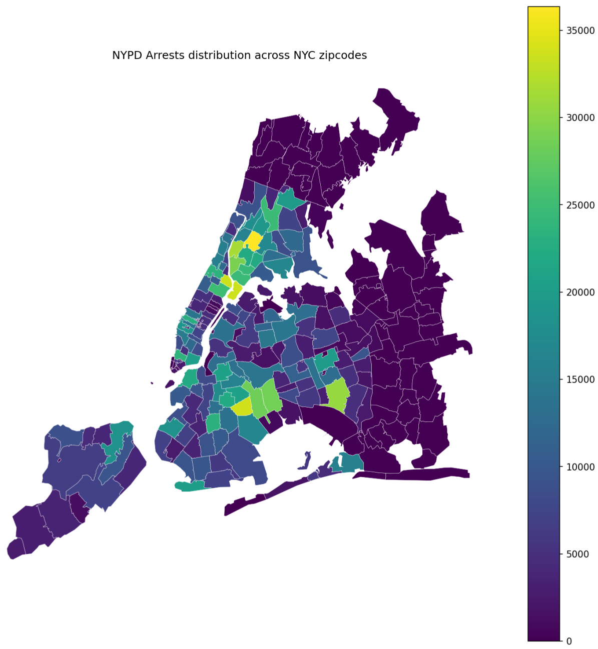

Fig. 4.1: NYPD Arrests distribution across NYC zipcodes.

Spatial distribution of arrests. The distribution of NYPD arrests exhibits a strong distribution across NYC zipcodes (Figure 4.1). The arrests are highly concentrated in a small number of zipcodes like Jamaica and the Bronx, while the majority of zipcodes appear in darker purple or blue shades, which show low arrest counts.

Borough-level patterns. At the borough level, Upper Manhattan and the South Bronx show the highest arrest rate, indicated by bright yellow and green regions. Central Brooklyn also contains zipcodes like Jamaica and neighbouring areas with elevated arrest counts.

Staten Island shows lower arrest levels relative to other boroughs, and Eastern Queens appears lighter, suggesting lower arrest density. These patterns align with known variations in population density across New York City.

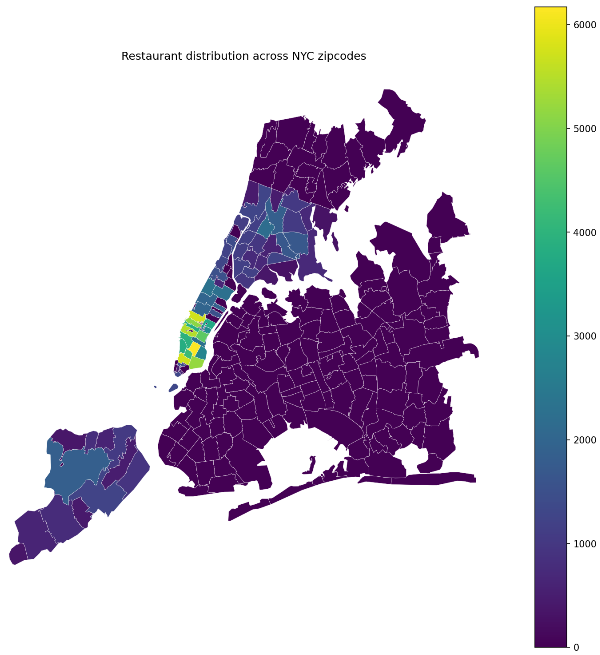

Fig. 4.2: Restaurants distributions across NYC zipcodes.

Spatial distribution of restaurants. According to our data, the restaurant distribution has most of its data points concentrated around Manhattan, with some in Staten Island (Figure 4.2). It is highly uneven and exhibits a strong spatial concentration. Most zipcodes appear in darker shades, indicating relatively low or no restaurant counts, while a small number of zipcodes are highlighted in yellow or green, corresponding to very high restaurant densities.

Borough-level patterns. At the borough level, restaurant density is heavily concentrated in Lower and Midtown Manhattan, particularly in commercial and zipcodes usually visited zipcodes. Outside Manhattan, restaurant counts decrease sharply, with portions of Brooklyn, Queens, the Bronx, and Staten Island showing comparatively low densities. This can be because of the reason that our dataset doesn't have enough datapoints outside Manhattan.

Spatial distribution of hotels. As observed in the hotel review distribution, the dataset contains a limited number of ZIP code data points outside Manhattan (Figure 4.3). As a result, the corresponding plot provides relatively little spatial information compared to other datasets. Nevertheless, it offers some insight, notably showing a higher concentration of hotel-related data points in Midtown Manhattan.

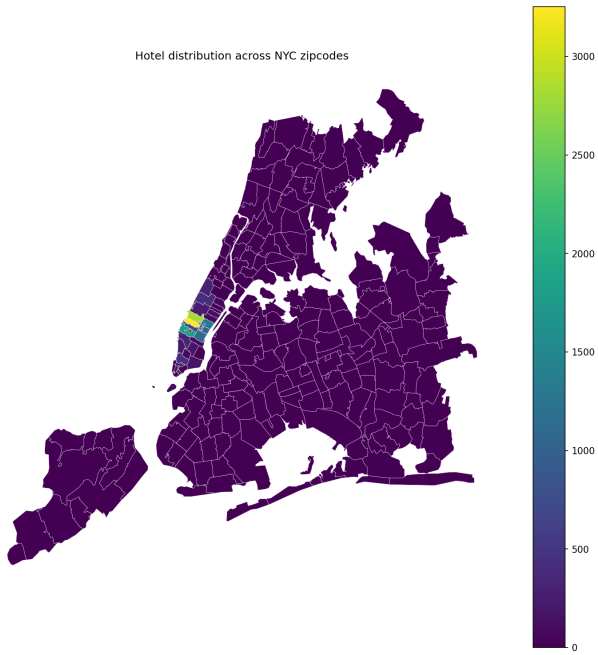

Fig. 4.3: TripAdvisor hotel distribution across NYC zipcodes.

During the MapReduce processing stage, we attempted to extract a broader range of ZIP codes however, we were unable to reliably parse sufficient data outside Manhattan. The lack of data points in outer boroughs limits our ability to analyze hotel distribution patterns beyond Manhattan.

Spatial distribution of MTA Rides. The distribution of total MTA rides across NYC zipcodes exhibits a strong spatial distribution, with a small number of zipcodes that result in very high ride volumes (Figure 4.4). While many ZIP codes appear in lower intensity ranges, the overall distribution is broader than that observed for restaurant or hotel datasets, indicating more widespread activity across the city.

Borough-level patterns. At the borough level, Central Brooklyn and parts of the Bronx show the highest concentrations of MTA rides, while Manhattan shows moderate to high ride volumes across many zipcodes. In contrast, much of Queens and Staten Island displays lower ride counts, with Staten Island remaining particularly low due to limited subway infrastructure. Overall, the MTA distribution reflects the geographic reach of public transportation rather than commercial or tourism-driven clustering.

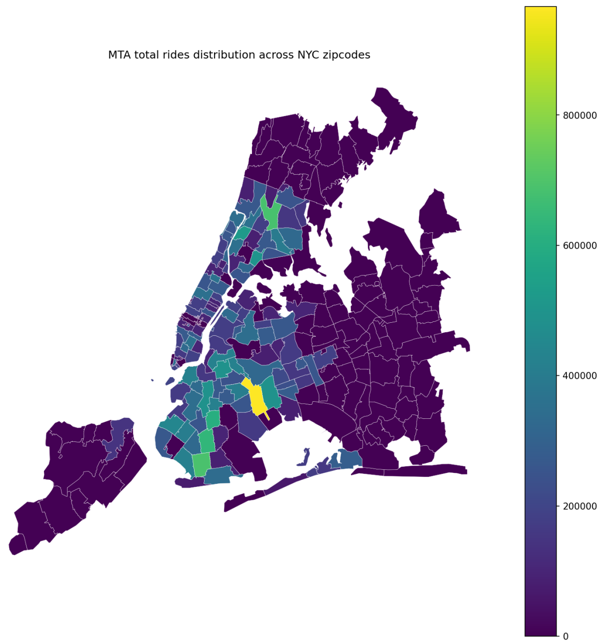

Fig. 4.4: MTA ridership distribution across NYC zipcodes.

4.2 Satisfaction index

The **Tourist Satisfaction Index (TSI)** is computed as a weighted composite score of four key components: hotel quality, food safety, public safety, and accessibility. The overall score is defined as:

$$\text{Score} = (H \times 0.30) + (F \times 0.20) + (S \times 0.30) + (A \times 0.20) \tag{4.1}$$

where:

- H **(Hotel Quality):**
$$H = \text{Average Rating} \times 2 \tag{4.2}$$
This scales the original 5-star hotel rating to a 10-point scale.

- F **(Food Safety):**
$$F = \frac{\text{\% Grade A Restaurants}}{10} \tag{4.3}$$
This converts the percentage of Grade A restaurants into a 10-point scale.

- S **(Safety Score):**
$$S = 10 - \min\left(\frac{\text{Total Arrests}}{50},\ 10\right) \tag{4.4}$$

This inverse scoring function ensures that higher crime rates result in lower safety scores.

- A **(Accessibility):**

$$A = \min\left(\frac{\text{Total Ridership}}{500{,}000},\ 10\right) \tag{4.5}$$

The score is capped at 10 to normalize extremely large transportation hubs.

Figure 4.5 illustrates the distribution of the satisfaction index across neighborhoods categorized by dominant cuisine type. Overall, cuisines such as American and Caribbean exhibit higher median satisfaction levels along with greater variability, suggesting that areas dominated by these cuisines tend to offer a broader mix of complementary amenities that influence tourist satisfaction. Chinese and Latin American cuisines display relatively tighter distributions around mid-range satisfaction values, indicating more consistent but moderate neighborhood experiences. In contrast, categories such as Coffee/Tea, Jewish/Kosher, and Mexican show limited dispersion, which may reflect more specialized neighborhood profiles with fewer contributing factors captured by the composite index. The presence of several high-value outliers, particularly within the American cuisine category, suggests that certain zip codes significantly outperform others in terms of overall satisfaction. These results indicate that dominant cuisine type can serve as a useful proxy for understanding variations in neighborhood-level tourist satisfaction, although substantial heterogeneity exists within cuisine groups.

While the preceding analysis highlights how satisfaction levels vary across dominant cuisine types, it is equally important to understand the prevalence of these cuisines across New York City neighborhoods in order to contextualize their overall influence. Figure 4.6 presents the distribution of the top dominant cuisines by number of neighborhoods, revealing that American cuisine overwhelmingly dominates the urban food landscape, followed by Caribbean, Chinese, and Latin American cuisines. This distribution helps explain why cuisines such as American and Caribbean play a prominent role in shaping aggregate satisfaction patterns observed earlier, as they characterize a substantially larger share of neighborhoods. In contrast, cuisines such as Coffee/Tea, Jewish/Kosher, and Greek appear in relatively few neighborhoods, suggesting that their narrower presence may contribute to the more specialized and less variable satisfaction profiles identified in the previous analysis. By examining cuisine prevalence alongside satisfaction outcomes, this figure reinforces the motivation for analyzing dominant cuisine as a neighborhood-level attribute and underscores its importance in interpreting broader trends in tourist satisfaction across the city.

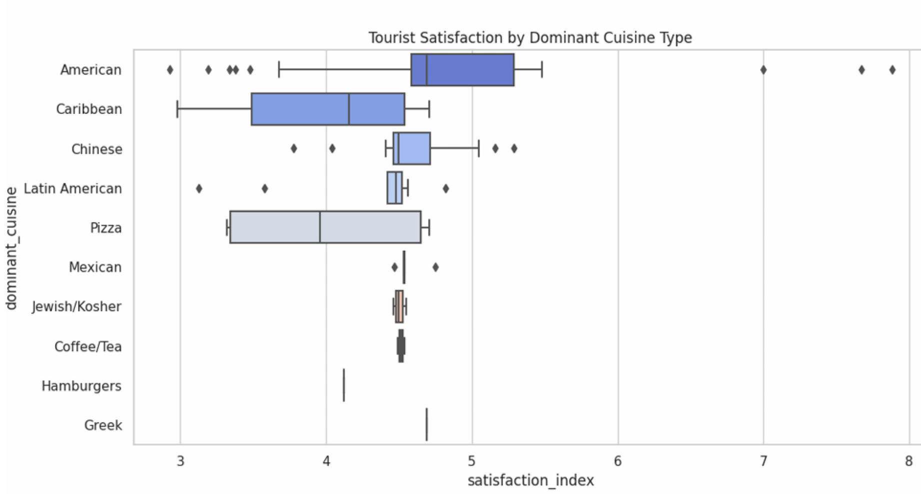

Fig. 4.5: TSI vs Dominant Cuisine

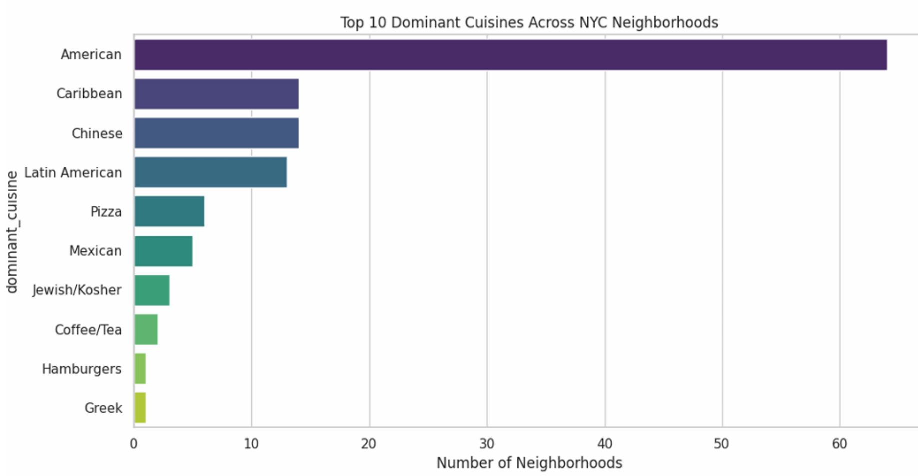

Fig. 4.6: Number of neighborhods vs Dominant cuisine

Figure 4.7 presents a borough-level comparison of average arrests, hotel quality, and overall satisfaction index. Manhattan stands out with the highest average hotel rating and satisfaction score, despite having moderately higher arrest counts than Queens, suggesting that strong lodging quality and urban amenities may offset safety concerns in shaping visitor perceptions. Queens exhibits the lowest average arrests alongside relatively high satisfaction, indicating a favorable balance between safety and visitor experience. In contrast, the Bronx and Staten Island show substantially higher arrest averages and lower satisfaction scores, highlighting potential challenges related to safety and perceived neighborhood appeal. Brooklyn occupies an intermediate position, with moderate arrest levels and satisfaction outcomes. Overall, the results suggest that while safety plays an important role in tourist satisfaction, hotel quality and broader urban infrastructure, particularly in Manhattan, significantly influence overall visitor experience at the borough level.

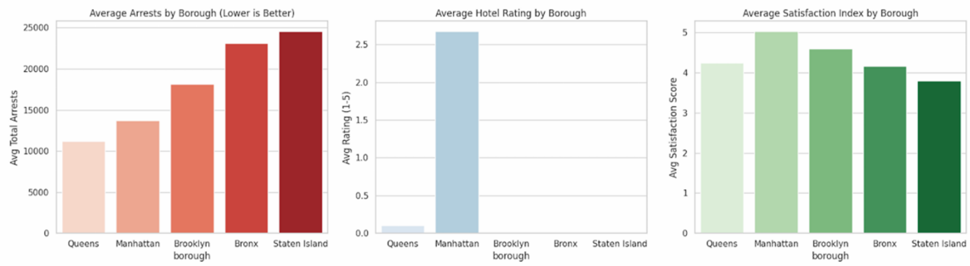

Fig. 4.7: Boroughwise Arrest, Hotel, TSI Analysis

Figure 4.8 finally examines the relationship between food quality, measured by the percentage of restaurants receiving a Grade A, and hotel quality, represented by average hotel ratings, with bubble size indicating total subway ridership and color denoting boroughs. Overall, neighborhoods with higher food quality tend to cluster around moderate to high hotel ratings, particularly in Manhattan, which also exhibits the largest bubbles, reflecting substantially higher transit usage. This suggests that areas combining strong dining standards and lodging quality also function as major transit hubs. In contrast, several zip codes across outer boroughs display relatively high food quality but near-zero hotel ratings, indicating limited hotel presence despite strong restaurant performance. A small number of outliers with low hotel ratings despite moderate food quality further highlight heterogeneity across neighborhoods. Taken together, the visualization suggests that while food quality and hotel quality are positively associated in core tourist areas, transit accessibility—captured through ridership—plays a critical role in distinguishing high-activity urban centers from residential or less tourism-oriented neighborhoods.

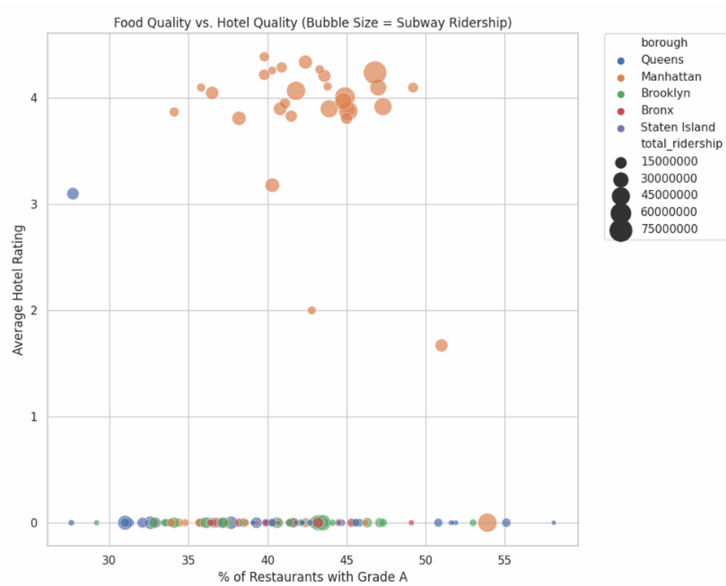

Fig. 4.8

Conclusion

This paper demonstrates a scalable big-data approach for integrating and analyzing heterogeneous urban datasets using Apache Hive and HDFS. By leveraging distributed storage and MapReduce-style query execution, we construct a unified, zip code–level representation of New York City neighborhoods that combines information on food quality, hotel ratings, crime activity, and subway ridership. The resulting Hive-based data model supports efficient aggregation and exploratory querying across large datasets that would be impractical to process using traditional relational systems.

Through HiveQL-driven aggregation, we generate neighborhood-level indicators that capture multiple dimensions relevant to tourism and urban activity, including safety, dining quality, lodging availability, transit usage, cuisine composition, and temporal mobility behavior. Borough- and cuisine-based comparisons reveal meaningful variation in satisfaction profiles and amenity distribution, while ridership-derived metrics such as the Leisure Intensity Score and Nightlife Activity Index provide additional temporal context that distinguishes commuter hubs from leisure- and nightlife-oriented areas.

Although the analysis presented is primarily descriptive, the integrated dataset and scalable processing pipeline establish a strong foundation for future work. Potential extensions include predictive modeling of neighborhood satisfaction, clustering and segmentation of urban areas using multi-dimensional features, and the incorporation of streaming or near–real-time data sources. Overall, this work highlights the effectiveness of Hive-based architectures for large-scale urban analytics and illustrates their applicability to tourism-oriented and city-scale data exploration tasks.

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Author Contributions

All authors contributed to the overall design of the study and the data processing framework. Each team member was responsible for profiling and cleaning one dataset using MapReduce. Aygun Najafova processed the NYC Restaurant Inspection dataset, Era Sarda worked on the TripAdvisor Hotel Review dataset, Debdeep Naha handled the MTA Ridership dataset, and Shivam Balikondwar processed the NYC Crime Complaint dataset. All authors reviewed and approved the final report. All source codes are available at [4]

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[4] *Tourist-Experience-Analysis-NYC.* https://github.com/Hyperion101010/nyc_tourist_trends_rbda.git.

[5] *TripAdvisor Hotel Review Data (via Kaggle).* https://www.kaggle.com/datasets/joebeachcapital/hotel-reviews/data. TripAdvisor.

Appendix

Column Name	Description	Data Type
ARREST_KEY	Unique persistent ID for each arrest	Text
ARREST_DATE	Exact date of arrest for the event	Timestamp
PD_CD	Three-digit internal classification code	Number
PD_DESC	Description corresponding to PD code	Text
KY_CD	Three-digit key code (more general classification)	Number
OFNS_DESC	Offense description based on KY code	Text
LAW_CODE	Legal code (NYS Penal Law, VTL, etc.)	Text
LAW_CAT_CD	Offense category: felony, misdemeanor, violation	Text
ARREST_BORO	Borough of arrest (B, S, K, M, Q)	Text
ARREST_PRECINCT	Precinct where arrest occurred	Number
JURISDICTION_CODE	Jurisdiction responsible (0–3, etc.)	Number
AGE_GROUP	Perpetrator age category (e.g., 25–44)	Text
PERP_SEX	Perpetrator sex	Text
PERP_RACE	Perpetrator race description	Text
X_COORD_CD	Midblock X-coordinate (NYS Plane Coordinate System)	Number
Y_COORD_CD	Midblock Y-coordinate (NYS Plane Coordinate System)	Number
Latitude	Geographic latitude of arrest location	Number
Longitude	Geographic longitude of arrest location	Number
Lon_Lat	Geographic point as “POINT(lon lat)”	Text

Table A.1: NYPD Arrests Dataset

Column_Name	**Description**	**Data_Type**
transit_timestamp	Timestamp when payment took place in local time.	DATE
transit_mode	Distinguishes between the subway, Staten Island Railway, and the Roosevelt Island Tram.	Text
station_complex_id	Unique alphanumeric identifier for subway station complexes.	ALPHANUMERIC
station_complex	The subway complex where an entry swipe or tap occurred.	Text
borough	Borough of New York City serviced by the subway system (Bronx, Brooklyn, Manhattan, Queens).	Text
payment_method	Specifies whether the payment method used was OMNY or MetroCard.	Text
fare_class_category	Class of fare payment used for the trip, including MetroCard and OMNY.	Text
ridership	Total number of riders entering a subway complex at the specified hour for the given fare type.	NUMERIC
transfers	Number of riders entering via free bus-to-subway or free out-of-network transfers.	NUMERIC
latitude	Latitude of the specified subway complex.	DECIMAL
longitude	Longitude of the specified subway complex.	DECIMAL

Table A.2: MTA Subway Hourly Ridership Dataset

Column Name	Description	Data Type
CAMIS	Unique restaurant identifier	Text
DBA	Restaurant name	Text
BORO	Borough where the restaurant is located	Text
BUILDING	Building number of the restaurant address	Text
STREET	Street name of the restaurant address	Text
ZIPCODE	ZIP code of the restaurant	Text
PHONE	Restaurant phone number	Text
CUISINE_DESCRIPTION	Category of cuisine offered by the restaurant	Text
INSPECTION_DATE	Date on which the inspection was conducted	Timestamp
ACTION	Outcome or action resulting from the inspection	Text
VIOLATION_CODE	Code identifying the type of violation	Text
VIOLATION_DESCRIPTION	Textual description of the violation	Text
CRITICAL_FLAG	Indicator of violation severity	Text
SCORE	Numerical score assigned during inspection	Number
GRADE	Inspection grade (A, B, C, etc.)	Text
GRADE_DATE	Date on which the grade was issued	Timestamp
RECORD_DATE	Date the record was created or last updated	Timestamp
INSPECTION_TYPE	Type or category of inspection conducted	Text
Latitude	Geographic latitude of the restaurant location	Number
Longitude	Geographic longitude of the restaurant location	Number
Community Board	Community board identifier (removed during preprocessing)	Number
Council District	City council district identifier (removed during preprocessing)	Number
Census Tract	Census tract identifier (removed during preprocessing)	Number
BIN	Building Identification Number (removed during preprocessing)	Number
BBL	Borough–Block–Lot identifier (removed during preprocessing)	Number
NTA	Neighborhood Tabulation Area code (removed during preprocessing)	Text
Location	Combined geographic location field	Text

Table A.3: NYC Restaurant Inspection Dataset

Column Name	Description	Data Type
title	Title of the review	Text
text	Full textual content of the review	Text
offeringId	Unique identifier for the hotel or offering being reviewed	Integer
numHelpfulVotes	Number of users who marked the review as helpful	Integer
reviewId	Unique identifier for the review	Integer
hotelClass	Star classification of the hotel	Float
regionId	Unique identifier for the geographic region	Integer
url	URL of the review or hotel page	Text
type	Type or category of the offering	Text
name	Name of the hotel or offering	Text
region	Region where the hotel is located	Text
locality	City or locality of the hotel	Text
streetAddress	Street address of the hotel	Text
zipcode	ZIP or postal code of the hotel location	Text
dateReviewed	Date when the review was submitted	Timestamp
dateStayed	Date when the reviewer stayed at the hotel	Timestamp
service	Rating score for service quality	Float
cleanliness	Rating score for cleanliness	Float
overall	Overall rating given by the reviewer	Float
valueRating	Rating score for value for money	Float
locationRating	Rating score for hotel location	Float
sleepQuality	Rating score for sleep quality	Float
rooms	Rating score for room quality	Float
fullAddress	Full textual address of the hotel	Text

Table A.4: Hotel Reviews Dataset: Offerings & Ratings Merged

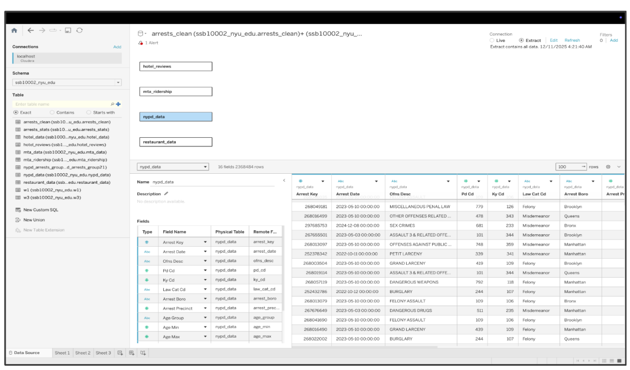

Fig. A.1: Tableau